

EE5311 Digital IC Design: Assignment 1

Transfer characteristics of transistors

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1. nMOS Transistor Characteristics

(a) Transfer Characteristics (I_{ds} vs V_{gs})

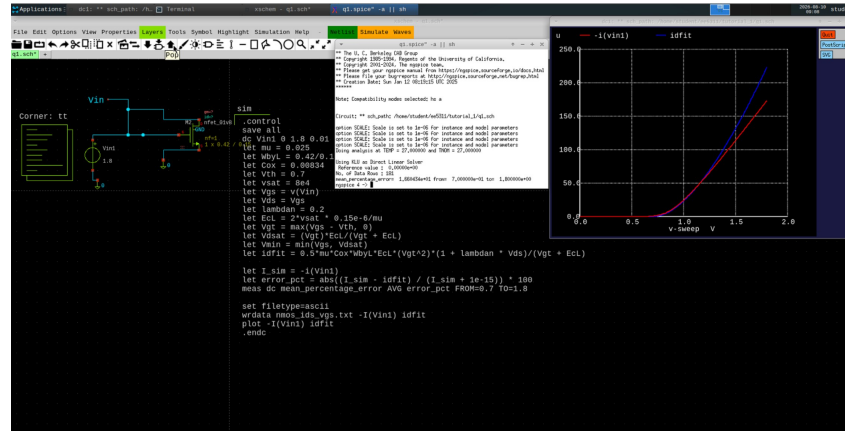


Figure 1: Simulated vs Analytical Transfer Characteristics with Mean Error.

Region of Operation:

Is the transistor in the saturation or linear region?

Since the gate is connected to the drain, $V_{ds} = V_{gs}$. Therefore, $V_{ds} > V_{gs} - V_{th}$ is always satisfied, keeping the transistor in saturation.

Mean Percentage Error Calculation:

Calculated mean error = 16.6%

(b) Output Characteristics (I_{ds} vs V_{ds})

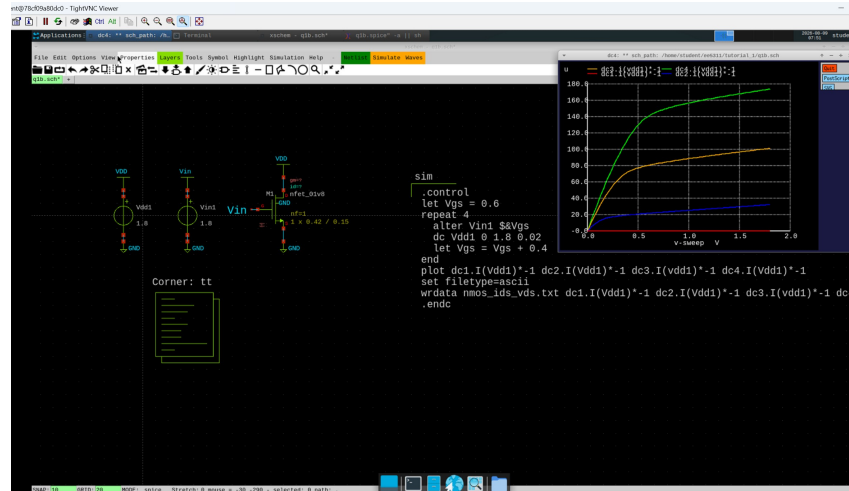


Figure 2: Simulated Output Characteristics.

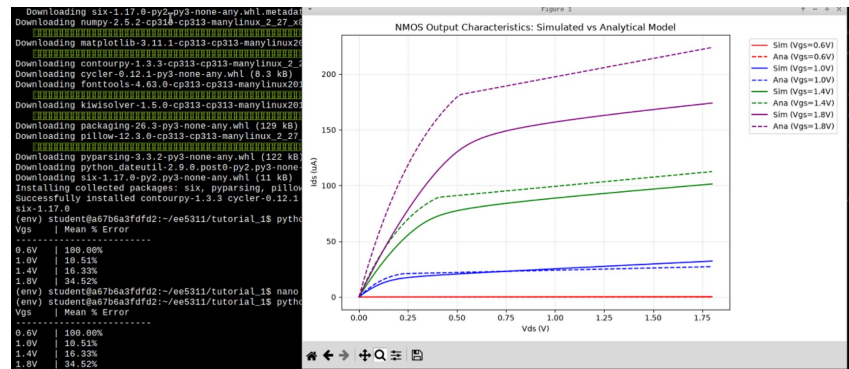


Figure 3: Mean Error.

Mean Percentage Error Calculation:

V_{gs}	Mean % Error
0.6V	100.00%
1.0V	10.51%
1.4V	16.33%
1.8V	34.52%

Table 1: Mean Percentage Error for varying V_{gs}

(c) Velocity Saturation Effect with Increasing Length

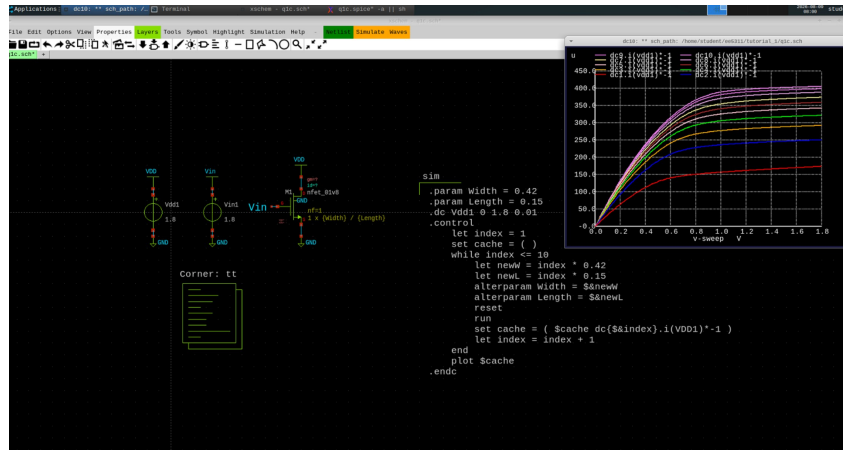


Figure 4: I_{ds} vs V_{ds} for varying W/L (constant ratio, increasing length).

2. pMOS Transistor Characteristics

(a) Transfer Characteristics

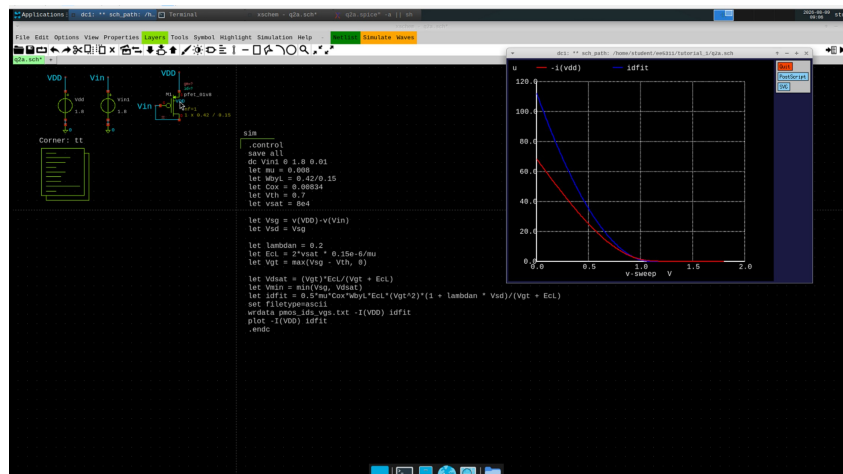


Figure 5: pMOS Simulated vs Analytical Transfer Characteristics.

Region of Operation: Saturation

(b) Output Characteristics

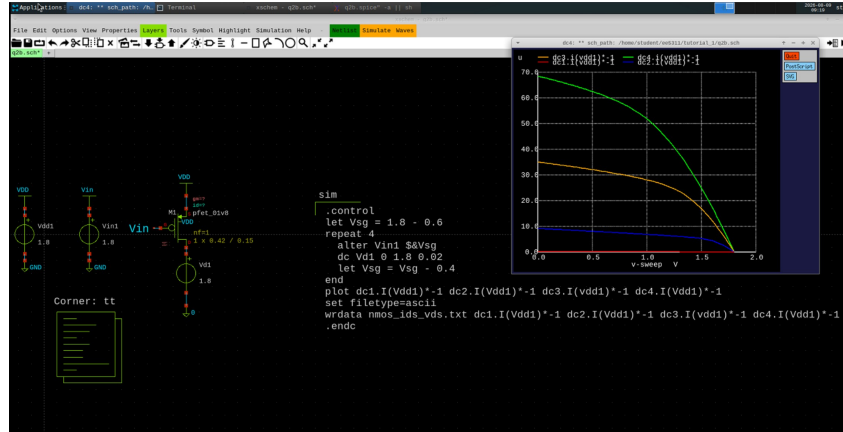


Figure 6: pMOS Output Characteristics.

(c) Velocity Saturation Effect with Increasing Length

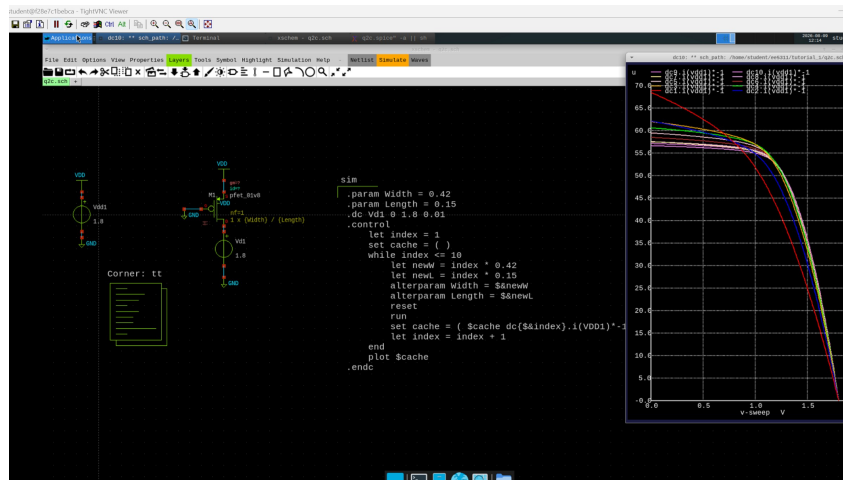


Figure 7: pMOS I_{ds} vs V_{ds} for varying W/L .

3. Transient Characteristics

(a) nMOS with Source Capacitor (Rising Step Input)

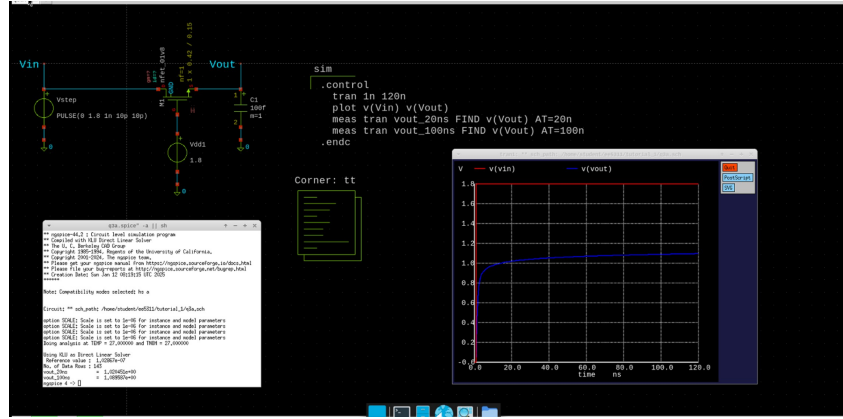


Figure 8: Transient response of nMOS with 100fF source capacitor (Step Input).

Simulated Values:

- V_{out} at $t = 20\text{ns}$: **1.02V**
- V_{out} at $t = 100\text{ns}$: **1.08V**

Analytical Comparison:

$$V_{out\infty} = (V_{DD} - V_{tn}) = 1.8 - 0.7 = 1.1V$$

(b) pMOS with Drain Capacitor (Rising Step Input)

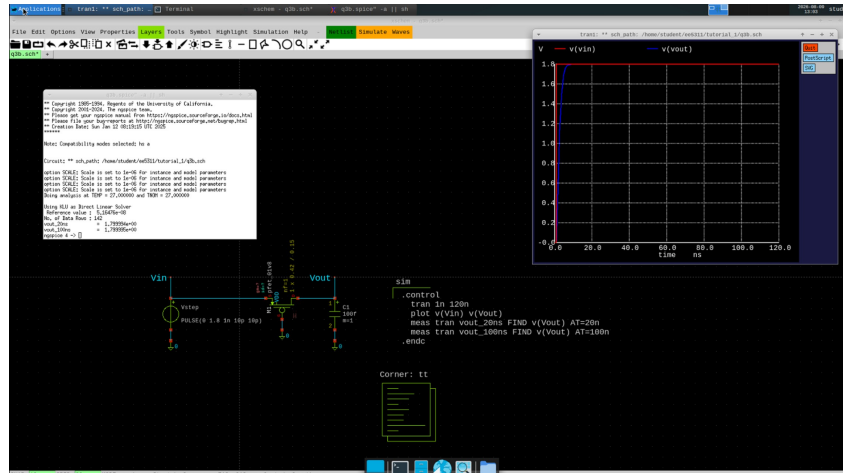


Figure 9: Transient response of pMOS with 100fF drain capacitor.

Simulated Values:

- V_{out} at $t = 20\text{ns}$: **1.799994V**
- V_{out} at $t = 100\text{ns}$: **1.799995V**

(c) nMOS with Source Capacitor (Falling Step Input)

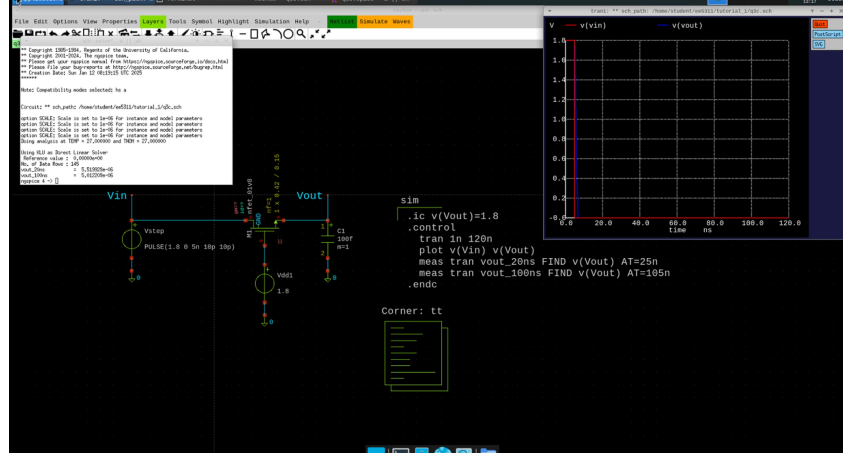


Figure 10: Transient response of nMOS for a falling waveform at V_{in} .

Simulated Values:

- V_{out} at $t = 20\text{ns}$: $5.5\mu\text{V}$
- V_{out} at $t = 100\text{ns}$: $5\mu\text{V}$

(d) pMOS with Source Capacitor (Falling Step Input)

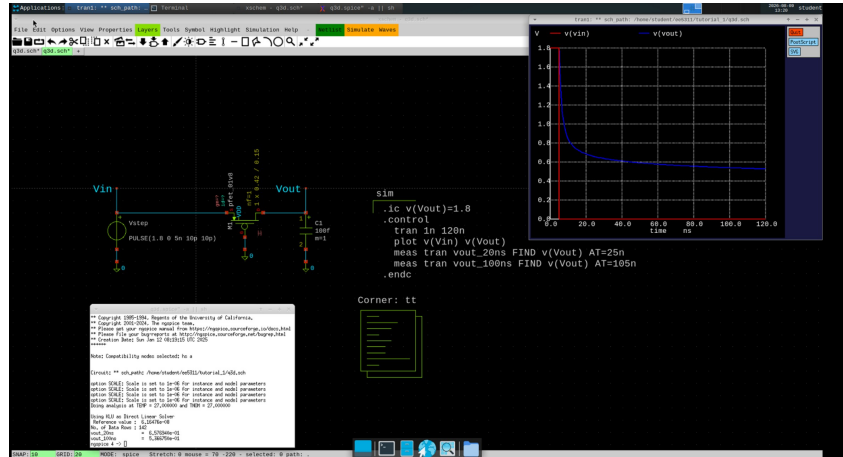


Figure 11: Transient response of pMOS for a falling waveform at V_{in} .

Simulated Values:

- V_{out} at $t = 20\text{ns}$: 0.657V
- V_{out} at $t = 100\text{ns}$: 0.536V

Analytical Comparison:

$$V_{out\infty} = (V_{tp}) = 0.7\text{V}$$